Q1 - 25 July - Shift 2

Two billiard balls of mass 0.05 kg each moving in opposite directions with 10 ms^{-1} collide and rebound with the same speed. If the time duration of contact is $t = 0.005$ s, then what is the force exerted on the ball due to each other?

(A) 100 N (B) 200 N
(C) 300 N (D) 400 N

Space for your notes:

Q2 - 25 July - Shift 2

Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to 3 m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}$ m. The value of x is

Space for your notes:

Q3 - 26 July - Shift 2

A ball of mass 0.15 kg hits the wall with its initial speed of 12 ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N. calculate the time duration of the contact of ball with the wall.

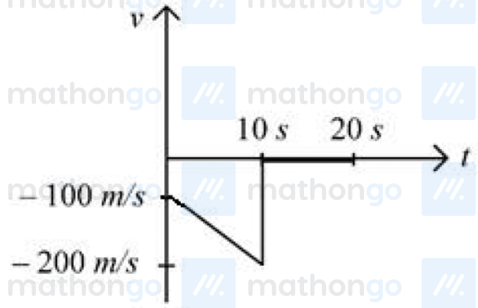
(A) 0.018 s (B) 0.036 s
(C) 0.009 s (D) 0.072 s

Space for your notes:

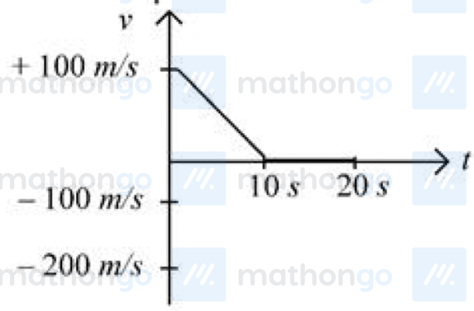
Q4 - 27 July - Shift 1

A bullet is shot vertically downwards with an initial velocity of 100 m/s from a certain height. Within 10 s, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time t = 20 s will be : (Take $g = 10\ m/s^2$)

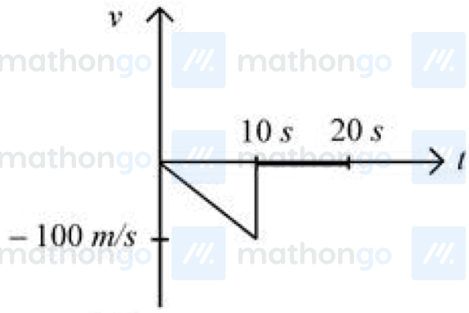
(A)



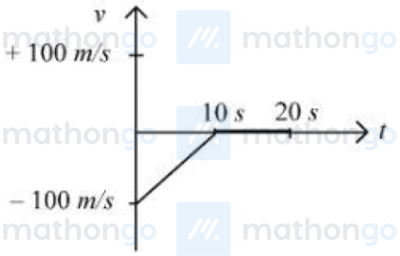
(B)



(C)



(D)



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Q5 - 27 July - Shift 2

The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at $(4 + x)$ cm inside the block. The value of x is:

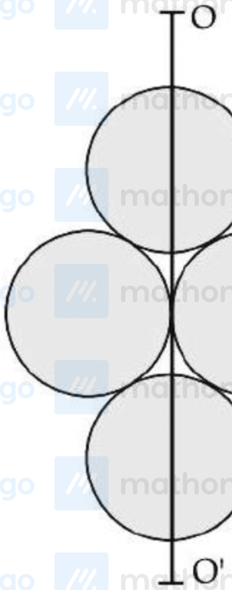
(A) 2.0 (B) 1.0

(C) 0. 5 (D) 1.5

Space for your notes:

Q6 - 28 July - Shift 1

Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}Ma^2$. Then, the value of x will be ______.



Space for your notes:

Q7 - 28 July - Shift 2

The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\rho = \rho_0 \left(1 - \frac{x^2}{L^2}\right)$ kg/m and length L (in meter) is $\frac{3L}{\alpha}$ m. The value of α is .... (where x is the distance form end A)

Space for your notes:

Q8 - 29 July - Shift 1

Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i} + 2\hat{j} + \hat{k}$ and $-3\hat{i} - 2\hat{j} + \hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

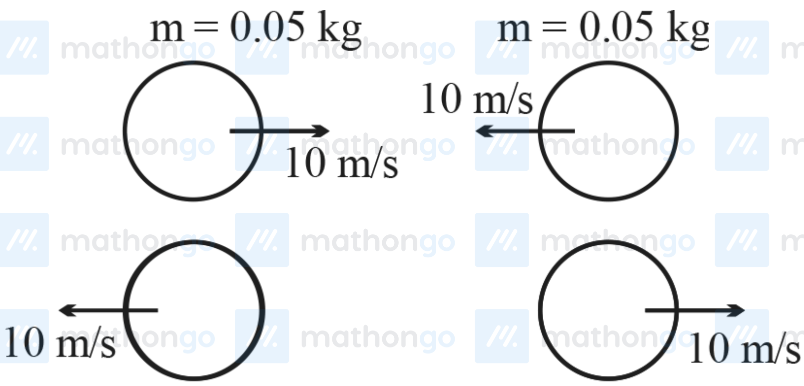
(A) $\hat{i} - 2\hat{j} + \hat{k}$
(B) $-3\hat{i} - 2\hat{j} + \hat{k}$
(C) $-2\hat{i} + 2\hat{k}$
(D) $-2\hat{i} - \hat{j} + 2\hat{k}$

Space for your notes:

Answer Key

Q1 (B)	**Q2** (2)	**Q3** (B)	**Q4** (A)
Q5 (C)	**Q6** (3)	**Q7** (8)	**Q8** (A)

Q1 (B)



Change in momentum of any one ball

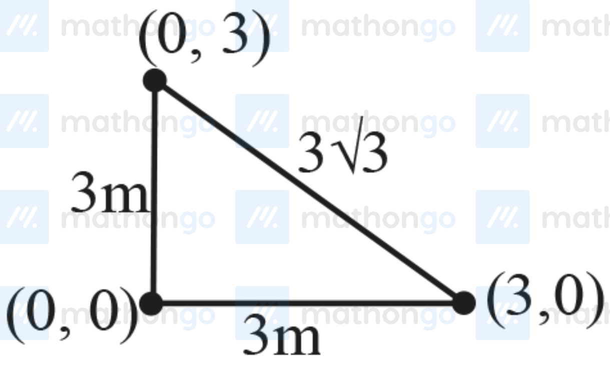
$$|\Delta \vec{P}| = 2 \times 0.05 \times 10$$

$$|\Delta \vec{P}| = 1$$

$$|\vec{F}_{av}| = \frac{|\Delta \vec{P}|}{\Delta t}$$

$$F_{av.} = 200 \text{ N}$$

Q2 (2)



$$\vec{r}_{com} = \frac{M(0\hat{i}+0\hat{j})+M(3\hat{i})+M(3\hat{j})}{3M}$$

$$\vec{r}_{com} = \hat{i}+\hat{j}$$

$$|\vec{r}_{com}| = \sqrt{2} = \sqrt{x}$$

$$x = 2$$

Q3 (B)

$$\vec{P}_i = 0.15 \times 12(\hat{i})$$

$$\vec{P}_f = 0.15 \times 12(-\hat{i})$$

$$|\overrightarrow{\Delta P}| = 3.6 \text{ kg} - \text{m/s}$$

$$3.6 = F\,\Delta t$$

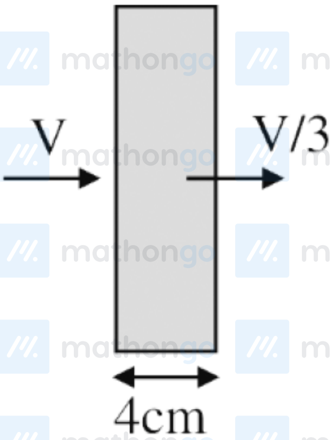
$$3.6 = 100\ \Delta t$$

$$\Delta t = 0.036 \text{ sec}$$

Q4 (A)

$$V = -100 - 10t$$

Q5 (C)



$$\left(\frac{V}{3}\right)^2 = V^2 - 2a(4) \Rightarrow a = \frac{8V^2}{9(8)} = \frac{V^2}{9}$$

$$0 = V^2 - 2a(4 + x)$$

$$\Rightarrow \quad V^2 = 2\left(\frac{V^2}{9}\right)(4 + x)$$

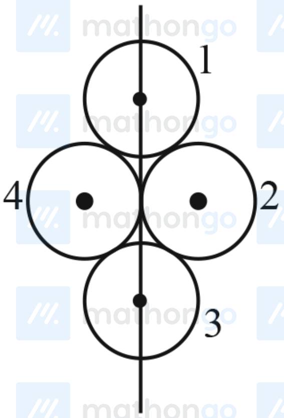
$$4.5 = 4 + x$$

$$x = 0.5$$

Q6 (3)

$$I_1 = I_3 = \frac{MR^2}{4}$$

$$I_2 = \frac{MR^2}{4} + MR^2 = \frac{5}{4}MR^2 = I_4$$



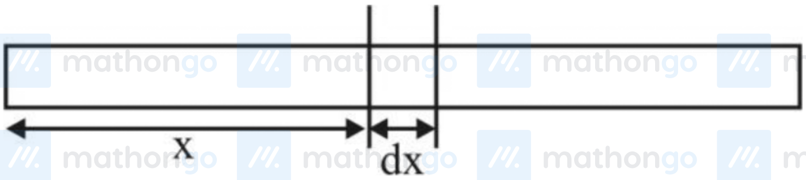
$$\text{So } I = I_1 + I_2 + I_3 + I_4$$

$$= \frac{MR^2}{2} + \frac{5}{2}MR^2$$

$$= 3MR^2, \text{ Putting } R = \frac{a}{2}$$

$$I = \frac{3Ma^2}{4}, \text{So } x = 3$$

Q7 (8)



$$dm = \lambda \cdot dx = \lambda_0\left(1 - \frac{x^2}{\ell^2}\right)$$

$$X_{cm} = \frac{\int x\,dm}{\int dm_\ell}$$

$$= \frac{\lambda_0 \int_0^\ell \left(x - \frac{x^3}{\ell^2}\right)dx}{\int_0^\ell \lambda_0\left(1 - \frac{x^2}{\ell^2}\right)dx} = \frac{\frac{\ell^2}{2} - \frac{\ell^4}{4\ell^2}}{\ell - \frac{\ell^3}{3\ell^2}} = \frac{3\ell}{8}$$

Q8 (A)

$$\overrightarrow{r_{com}} = \frac{m_1\vec{r_1} + m_2\vec{r_2}}{m_1 + m_2} = \frac{1(\hat{i} + 2\hat{j} + \hat{k}) + 3(-3\hat{i} - 2\hat{j} + \hat{k})}{1+3}$$

$$= -2\hat{i} - \hat{j} + \hat{k}$$

$$|2\hat{i} - \hat{j} + \hat{k}| = \sqrt{(2)^2 + (1)^2 + (1)^2} = \sqrt{6}$$